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On the modelling of the anhysteretic magnetization of homogeneous soft magnetic materials

Josefina María Silveyra *, Juan Manuel Conde Garrido

Universidad de Buenos Aires. Facultad de Ingeniería. Laboratorio de Sólidos Amorfos. Buenos Aires, Argentina
CONICET - Universidad de Buenos Aires. Instituto de tecnologías y Ciencias de la ingeniería "Hilario Fernández Long" (INTECIN). Buenos Aires, Argentina

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ABSTRACT

The analysis and optimization of devices with soft magnetic cores, such as inductive sensors and electric machines, rely on a simple and precise modelling of their nonlinear anhysteretic magnetization. The Langevin-Weiss equation of state has the advantages of being physically-based and capable of accurately describing isotropic materials. In fact, it has even been adopted as the backbone of the popular Jiles-Atherton model to describe magnetic hysteresis. However, many studies show the difficulty of retrieving the modelling parameters given an experimental anhysteretic curve, for which stochastic optimization methods are generally used.

In this work, we present a simple and deterministic method for finding the parameters of the Langevin-Weiss model: the saturation magnetization, the temperature dependent a parameter, and the molecular field constants. To this end, we optimize a problem with a nonlinear solver where the fitting variables are instrinsically bounded. We validated the approach by analyzing three widely used soft magnets; a grain-oriented silicon steel, a Mn-Zn soft ferrite, and a Fe-based nanocrystalline alloy with induced anisotropy.

We demonstrate that, contrary to what has previously been assumed, the Langevin-Weiss function can provide a good description of the anhysteretic magnetization of any homogeneous material, even materials with strong transverse uniaxial anisotropy. The key, which seems to have been overlooked, is to allow the Weiss molecular field to be negative.

1. Introduction

Soft magnetic materials play a key role in the conversion of energy throughout the world [1]. The need for highly efficient devices in the area of power electronics and electric machines result in a growing interest in the development of fast and reliable descriptions of the nonlinear magnetization process. The single-valued ideal, hysteresis-free magnetization curve, also called anhysteretic magnetization curve [2,3], is used by researchers focusing on better understanding the underlying physics of magnetization, as well as by engineers focusing on developing well-designed components with magnetic cores (sensors, transformers, electric motors, and other kinds of inductive devices) [4,5].

The anhysteretic magnetization would be observed if a material were successfully demagnetized by an alternating magnetic field together with a chosen steady magnetic field. The amplitude of the alternating field should be initially large enough to magnetically saturate the sample in both directions and effectively wipe out the previous magnetic history. Then, the amplitude should be exponentially decreased until only the steady field remains. The entire curve would be built up by varying the steady field. However, direct measurements are rarely reported in the literature, as they are tedious and time-consuming [6–9]. The alternative method of obtaining the anhysteretic curve by cooling down the sample from above the Curie point under various steady fields, is not only tedious but materials may suffer structural transformations along the heating process. Still, it is not clear if they indeed produce true anhysteretic curves, as even its physical description remains under discussion: one approach defines the anhysteretic curve as the magnetization that would occur in the absence of the material imperfections, while another approach defines it as the thermodynamic equilibrium state [10]. Due to these technical challenges, it is a common practice to approximate the anhysteretic magnetization as the average curve of the ascending and descending branches of the major hysteresis loop [11,12].

Despite the discrepancy on the physical definition of the term, the nonlinear anhysteretic magnetization curve is widely used to describe, as first approximation, the magnetic behavior of soft magnetic materials

* Corresponding author.
E-mail address: jsilveyra@fi.uba.ar (J.M. Silveyra).

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when modelling magnetic circuits. Also, the anhysteretic magnetization curve is the basis of the Jiles-Atherton model of magnetic hysteresis [13,14], which is broadly used in modern lumped-element modeling (SPICE, Simulation Program with Integrated Circuits Emphasis, e.g. since Micro-Cap v. 8 released in 2004 [15]) and finite element analysis software (e.g., since COMSOL Multiphysics® v. 5.2a released in 2016 [16]).

In this work, we asked whether it is possible to describe the anhysteretic curve of any homogeneous soft magnet with an equation of state easy to implement, yet accurate and physically-based.

2. Langevin's and Weiss' theories of magnetism

Several empirical formulae have been proposed to model the anhysteretic magnetization curve, including rational, logarithmic, exponential, and inverse tangent functions, among others [2,17]. Phenomenological approaches have the advantage of generality, but typically provide less insight into the physics of the problem than models developed from physical principles [18]. Here, we will focus on the most popular equation, the semi-theoretical Langevin-Weiss function because of its remarkable physical simplicity and yet being successful in describing the salient facts of ferromagnetism. In the following, we will briefly review the theories of Langevin and Weiss.

The Langevin model of paramagnetism, proposed in 1905 [19], treats the material as an assembly of elementary entities in thermodynamic equilibrium that do not interact with one another. All magnetic entities carry the same permanent magnetic dipole moment $\boldsymbol{m_d}$, whose distribution-in-angle follows the Boltzmann distribution (the theory is semi-classical or semi-quantum in nature, as Langevin assigned a magnetic moment to the elementary entities that can point in any direction, but with a fixed value, quantizing thus the system without recognizing it explicitly [20,21]). In the absence of an applied field, due to thermal agitation, the magnetic moments are randomly oriented cancelling one another out, so that the magnetization of the paramagnet is zero. When a magnetic field is applied, there is a static equilibrium between the thermal agitation and the tendency of the magnetic moments to align in the same direction as the external field. For a magnetic moment oriented an angle θ from a magnetic field $\boldsymbol{H}$, the probability p of having an energy $E(\theta)$ at a temperature T is, according to the Boltzmann statistics:

$$p(E) \propto e^{-\frac{E}{k_B T}} \tag{1}$$

where k_B is the Boltzmann constant and T is the absolute temperature. The potential energy of a magnetic moment depends upon its orientation with respect to the magnetic field. When a magnetic moment $\boldsymbol{m_d}$ is placed in an external magnetic field $\boldsymbol{H}$, it experiences a torque given by:

$$\boldsymbol{\tau} = \boldsymbol{m_d} \times \mu_0 \boldsymbol{H} \tag{2}$$

which tends to align it in the direction of the magnetic field (μ_0 is the vacuum magnetic permeability). The work required to turn it through an angle $d\theta$ against the field is:

$$\delta W = \tau d\theta = \mu_0 m_d H \sin(\theta) d\theta \tag{3}$$

increasing the magnetostatic potential energy of the system. Choosing the potential energy to be zero at $\pi/2$, the energy of interaction between the field and the elementary magnetic moment (oriented at an angle θ from the field) is:

$$E(\theta) = W = \int_{\frac{\pi}{2}}^{\theta} \mu_0 m_d H \sin(\theta) d\theta = \mu_0 m_d H(-\cos(\theta)) \tag{4a}$$

$$E(\theta) = -\mu_0 \boldsymbol{m_d} \bullet \boldsymbol{H} \tag{4b}$$

The configuration is at stable equilibrium when $\boldsymbol{m_d}$ is aligned parallel to $\boldsymbol{H}$, minimizing the magnetostatic potential energy, and at unstable equilibrium when it is antiparallel.

For N magnetic moments per unit volume, the total magnetic moment per unit volume at thermal equilibrium and in the direction of the applied field, that is, the average magnetization of the sample M, is proportional to the thermal statistical average of $\cos(\theta)$:

$$M = N m_d \langle \cos(\theta) \rangle = N m_d \frac{\int_{E_{min}}^{E_{max}} \cos(\theta) p(E) dE}{\int_{E_{min}}^{E_{max}} p(E) dE} = N m_d \frac{\int_{E_{min}}^{E_{max}} \cos(\theta) e^{-\frac{E}{k_B T}} dE}{\int_{E_{min}}^{E_{max}} e^{-\frac{E}{k_B T}} dE} \tag{5}$$

where $m_d \cos(\theta)$ is the contribution of each of the N magnetic elementary entities per unit volume to the magnetization along the field direction. By substituting Eq. (4a) into (5), we get,

$$\begin{aligned} M &= N m_d \frac{\int_0^{\pi} \cos(\theta) e^{\frac{\mu_0 m_d}{k_B T} H \cos(\theta)} d(-\mu_0 m_d H \cos(\theta))}{\int_0^{\pi} e^{\frac{\mu_0 m_d}{k_B T} H \cos(\theta)} d(-\mu_0 m_d H \cos(\theta))} \\ &= N m_d \frac{\int_0^{\pi} \cos(\theta) e^{\frac{\mu_0 m_d}{k_B T} H \cos(\theta)} \sin(\theta) d\theta}{\int_0^{\pi} e^{\frac{\mu_0 m_d}{k_B T} H \cos(\theta)} \sin(\theta) d\theta} \end{aligned} \tag{6}$$

Notice that the limit inferior in Eq. (5), E_{min}, corresponds to $\theta = 0$ in Eq. (6), while the limit superior, E_{max}, corresponds to $\theta = \pi$. The quantity $N m_d$ is the maximum possible moment the material can have, which corresponds to the perfect alignment of all the moments, i.e., the saturation magnetization M_S. Then, by calling:

$$a = \frac{k_B T}{\mu_0 m_d} \tag{7}$$

we get,

$$M = M_S \frac{\int_0^{\pi} e^{\frac{H}{a} \cos(\theta)} \sin(\theta) \cos(\theta) d\theta}{\int_0^{\pi} e^{\frac{H}{a} \cos(\theta)} \sin(\theta) d\theta} \tag{8}$$

The above integrals lead to the so-called Langevin function $\mathscr{L}$:

$$\begin{aligned} M &= M_S \frac{\left[2\left(\frac{H}{a}\right)\cosh\left(\frac{H}{a}\right) - 2\sinh\left(\frac{H}{a}\right)\right] \Big/ \left(\frac{H}{a}\right)^2}{2\sinh\left(\frac{H}{a}\right) \Big/ \left(\frac{H}{a}\right)} \\ &= M_S \left[\frac{2\left(\frac{H}{a}\right)\cosh\left(\frac{H}{a}\right)}{\left(\frac{H}{a}\right)2\sinh\left(\frac{H}{a}\right)} - \frac{2\sinh\left(\frac{H}{a}\right)}{\left(\frac{H}{a}\right)2\sinh\left(\frac{H}{a}\right)}\right] = M_S \left[\frac{\cosh\left(\frac{H}{a}\right)}{\sinh\left(\frac{H}{a}\right)} - \frac{1}{\left(\frac{H}{a}\right)}\right] \\ &= M_S \left[\coth\left(\frac{H}{a}\right) - \left(\frac{a}{H}\right)\right] = M_s \left[\frac{e^{H/a} + e^{-H/a}}{e^{H/a} - e^{-H/a}} - \left(\frac{a}{H}\right)\right] = M_s \mathscr{L}\left(\frac{H}{a}\right) \end{aligned} \tag{9}$$

The Langevin function can also be written in terms of the reduced fields as:

$$\mathscr{L}(h) = m = \coth(h) - \frac{1}{h} \tag{10}$$

where $m = M/M_S$ is the reduced magnetization and $h = H/a$ is the reduced magnetic field.

For low fields (typically $|h| < 0.1$), the Langevin function can be approximated by a truncation of its Taylor series expansion about the origin (i.e., the Maclaurin series):

$$\mathscr{L}(h) = \frac{h}{3} - \frac{1}{45}h^3 + \frac{2}{945}h^5 - \frac{1}{4725}h^7 + \cdots \tag{11}$$

In the vicinity of $h = 0$, using only the first term of the Taylor series expansion, the Langevin function can be approximated to be:

$$\mathscr{L}(h) = m \approx \frac{h}{3} \tag{12}$$

which is numerically better than the direct evaluation of the actual analytical expression, since the Langevin function suffers from loss of significance.

In the above theory of paramagnetism, we assumed that the magnetic moment can take any orientation. If we allow the system to have only discrete directions, we obtain the so-called Brillouin function, compatible with quantum mechanics, which converges to the semi-classical Langevin function in the limit of the total angular momentum $J \to \infty$ [22] (in the other limit case, for a single electron spin with only two possible orientations, $J = 1/2$ and the Brillouin function reduces to the hyperbolic tangent function).

To explain the behavior of ferromagnetism, Weiss proposed in 1907 taking the effective field acting on the elementary magnets not to be the external field H, but rather this field augmented by a mean uniform internal field [23]:

$$H_{eff} = H + H_{int} \tag{13}$$

as an analogue to the internal pressure earlier proposed by Van der Waals in the modeling of the behavior of fluids [24]. Weiss proposed that this internal field, which he called molecular field, should be directly proportional to the average magnetization of the sample, dispensing the need of considering the spatial distribution of the magnetic moments:

$$H_{int} = \alpha M \tag{14}$$

where α is a proportionality constant independent of temperature (the demagnetizing factor is considered as a constant contribution to α [25]). For some cases, it would be necessary to expand the latter into higher order odd terms of M [26] (which still guarantees that the magnetization curve is antisymmetric with respect to the magnetic field):

$$H_{int} = \alpha_1 M + \alpha_3 M^3 + \cdots \tag{15}$$

With Weiss' modification, considering only the first term of the molecular field development, the reduced magnetic field becomes:

$$h = \frac{H_{eff}}{a} = \frac{H + \alpha M}{a} \tag{16}$$

and the Langevin-Weiss equation of state, which describes the dependence of the magnetization M on the external magnetic field H at a temperature T, results:

$$M(H) = M_s \mathscr{L}(h) = M_s \mathscr{L}\left(\frac{H + \alpha M}{a}\right) \tag{17}$$

Weiss was unable to explain the physical nature of the molecular field, as the magnetic dipole–dipole interactions were too weak to lead to ferromagnetic alignment. Dorfman provided experimental evidence of its non-magnetic character in 1927 [27]. In 1928, Heisenberg proposed a scenario of localized spins in which Coulomb interactions together with the Pauli exclusion principle, were sufficient to produce the same effects as the apparently large molecular field postulated by Weiss [28]. This quantum mechanical phenomenon is known as the exchange interaction between electronic spins and has no classical analogue. Under some considerations, the molecular field constant may be related to the exchange integral J_0, the main parameter of the exchange interaction, by:

$$\alpha = \frac{2zJ_0}{Ng^2\mu_B^2} \tag{18}$$

where z is the number of nearest neighbors (with which interaction occurs), N is the number of magnetic elementary entities per unit volume, g is the Landé spectroscopic splitting factor, and μ_B is the Bohr magneton [29]. For ferromagnetism to occur in Heisenberg's model, J_0 and hence α must be positive, provided that the principal quantum number is sufficiently high; $n \gtrapprox 3$ (if J_0 were negative, it would be favorable for the nearest neighbor magnetic moments to lie antiparallel to each other, i.e., the material would exhibit antiferromagnetic behavior). But the Heisenberg model is too simple for a quantitative description of real ferromagnetic materials, where orbital wave functions deform from the free atomic state. Among the inconsistences present in the theory, is the sign problem of J_0 [30]. In fact, there is no single example of a real ferromagnet for which it has been theoretically proven that J_0 must be positive [21].

3. Determination of the Langevin-Weiss model parameters: A state of art

The Langevin-Weiss equation of state cannot be solved analytically for any field, either for M (M must be found iteratively in Eq. (17) , as M depends on both H and M) or for H (the exact inverse of the Langevin function has no analytic representation). Also, its applicability is limited by the difficulty of retrieving the parameters M_S, a, and α given an experimental magnetization curve. Jiles et al. argued that M_S is often known from data sheets or other references [31], but that is not the case for new alloys synthetized by materials scientists. Even the properties of known commercial materials may vary from those shown in data sheets [32]. Also, magnetic measurements at arbitrarily high applied fields may not be readily available. The a parameter also presents a problem since, although it is certainly temperature dependent ($a = \frac{k_B T}{\mu_0 m_d} = \frac{N k_B T}{\mu_0 M_S}$), it does not simply represent the Boltzmann energy unless we assume a regular array of pseudo-domains (i.e., domains of equal magnetic moments and fixed size). Affected by anisotropy and texture, it varies along different directions in a textured material [31]. Finally, the α parameter is also hard to know in advance for real macroscopic materials [33].

In the curve fitting process, the objective is to minimize the sum of squared differences between each actual data value and the corresponding predictive value. The classic iterative procedure is very sensitive to the initial values of parameters and to the order of their evaluation [34]. The reason is that the optimization problem of the varying Langevin-Weiss model parameters (M_S, a, and α) presents local minima. Conversely, stochastic optimization methods using random variables and/or iterates in a trusted solution space is a challenging and time-consuming numerical task that may suffer from instability hazards. Moreover, it is not trivial how to define the variables bounds. M_s is obviously partially bounded with a known minimum: the magnetization measured at the tip of the experimental magnetization curve (M_{TIP}). The upper bound, sometimes set to be a 20 % larger than the latter ($1.2 \times M_{TIP}$) [35] to limit computation costs, may leave out the global solution in some cases (as in the grain-oriented electrical steel example analyzed later in this work). Bounds for a and α are usually set in wide –positive– ranges, yet differing by orders of magnitude for different materials [33]. In fact, after suggesting bounds for the three parameters (relating them to M_{TIP}, to the corresponding field H_{TIP}, and to the coercive field H_C of the hysteresis loop), Chwastek and Szczygłowski have recognized that, in specific cases, "some trial-and-error adjustments may be needed" [36].

Although stochastic optimization methods usually give good results for isotropic ferrites [37] and soft magnetic composite materials [25], they commonly fail in the case of anisotropic materials, as reported by Nowicki et al. [38]. For example, in Fig. 6c) of that study, the simulated curve of a nanocrystalline alloy with induced transverse anisotropy, optimized using a differential evolution algorithm, exhibited a peak induction 11 % lower and an initial permeability 2.7 times larger than the actual magnetization curve. These discrepancies have been ascribed to the Langevin-Weiss theory itself, rather than to the fitting process. Thus, interpolated curves [11] or more sophisticated and numerically expensive models [7,39] have been proposed to describe the anhysteretic magnetization. For example, Ramesh et al. have proposed a model for materials with uniaxial anisotropy that needs to be numerically integrated. The model reduces to the Langevin-Weiss function for the case of no anisotropic energy (after a typo correction reported by Szewczyk), so it should always give a solution with an equal or better goodness of fit R^2 than the Langevin-Weiss function. However, the results reported by

Nowicki et al. show the contrary for an amorphous alloy with parallel anisotropy (see Tables 4 and 5 in [38]), demonstrating the intrinsic uncertainty of stochastic optimization strategies.

One more issue related to stochastic techniques, is that the optimally fitted solution may result in an unphysical magnetization curve. Several researchers have discussed the problem of retrieving a set of parameters that describes an anhysteretic magnetization curve with an unphysical negative initial magnetic susceptibility (χ_{in}) and thus lying in all four quadrants of the $M-H$ plane, rather than in the first and third quadrants only [17,40–42]. This occurs especially when fitting steep curves, as those from typical grain-oriented steels in the rolling direction [40]. In terms of the Langevin-Weiss model, the anhysteretic susceptibility at the origin is given by (see Eqs. (12) and (17)):

$$\chi_{in} = \left.\frac{\partial M}{\partial H}\right|_{H,M\to 0} = \left.\frac{\partial M_S \mathscr{L}(h)}{\partial H}\right|_{H,M\to 0} \cong M_S \left.\frac{\partial\left(\frac{h}{3}\right)}{\partial H}\right|_{H,M\to 0} = \frac{M_S}{3}\left.\frac{\partial\left(\frac{H+\alpha M}{a}\right)}{\partial H}\right|_{H,M\to 0}$$

$$\chi_{in} = \frac{M_S}{3a}(1+\alpha\chi_{in}) \tag{19}$$

and so,

$$\chi_{in} = \frac{M_S}{3a-\alpha M_S} \tag{20}$$

Therefore, from the condition of positive χ_{in}, we may find a relative upper bound on the molecular field constant α to implement in the optimization process and avoid modelling unphysical curves:

$$\alpha < \frac{3a}{M_S} \tag{21}$$

4. A new method for the determination of the Langevin-Weiss model parameters

To overcome the challenges described above that are intrinsic to stochastic optimization strategies, we report a simple, computationally efficient, and yet accurate method for retrieving the three parameters of the Langevin-Weiss equation, given an anhysteretic magnetization curve. In this approach, the problem of optimizing three parameters with uncertain bounds, is transformed into a problem of two -or even one- parameter with defined bounds that tackles the local minima problem. To this end, we solve a system of equations with the three unknowns –M_S, a, and α– and three restrictions. First, we constrain the predictive curve to pass through the tip of the data curve (i.e., the point measured at the maximum external field); (H_{TIP}, M_{TIP}). The second constraint is any other point of the experimental curve; (H_X, M_X), which may be either optimized for obtaining a better fit, or strategically selected for obtaining a faster solution. Notice that the zero point $(H, M) = (0,0)$ is already a point of the predicted anhysteretic curve with the Langevin equation (it corresponds to the demagnetized state that implies that the distribution of the elementary magnetic moments is completely uniform in all directions [3]). The third and last condition arise from minimizing the objective function; the mean-squared-error of the fitted curve. Below, we describe the algorithm of the method.

Step 1: Propose a tip reduced magnetization, m_{TIP}, which must be positive and below unity (in Step 7, the optimum m_{TIP} will be found).

Step 2: Determine the saturation magnetization by:

$$M_S = m_{TIP} M_{TIP} \tag{22}$$

where M_{TIP} is the tip magnetization of the data curve to fit.

Step 3: Choose the second constraint magnetization, M_X, which must be positive and below M_{TIP} (again, in Step 7, the optimum M_X will be found). Evaluate its reduced value by:

$$m_X = M_X/M_S \tag{23}$$

Step 4: Estimate the corresponding magnetic field H_X by interpolation of the data curve.

Step 5: Calculate a and α by solving the system of two equations that arise from evaluating Eq. (9) at the two constraint points:

$$m_{TIP} = \mathscr{L}(h_{TIP}) = \mathscr{L}\left(\frac{H_{TIP}+\alpha M_{TIP}}{a}\right) \tag{24a}$$

$$m_X = \mathscr{L}(h_X) = \mathscr{L}\left(\frac{H_X+\alpha M_X}{a}\right) \tag{24b}$$

which may be rearranged as follows:

$$a = \frac{H_{TIP}+\alpha M_{TIP}}{\mathscr{L}^{-1}(m_{TIP})} \tag{25a}$$

$$a = \frac{H_X+\alpha M_X}{\mathscr{L}^{-1}(m_X)} \tag{25b}$$

and then equated and solved for the molecular field constant α:

$$\alpha = \frac{H_{TIP}}{M_{TIP}}\frac{\frac{\mathscr{L}^{-1}(m_X)}{\mathscr{L}^{-1}(m_{TIP})}-\frac{H_X}{H_{TIP}}}{\frac{m_X}{m_{TIP}}-\frac{\mathscr{L}^{-1}(m_X)}{\mathscr{L}^{-1}(m_{TIP})}} = \frac{H_{TIP}}{M_{TIP}}\frac{\frac{h_X}{h_{TIP}}-\frac{H_X}{H_{TIP}}}{\frac{m_X}{m_{TIP}}-\frac{h_X}{h_{TIP}}} \tag{26}$$

Finally, a can be evaluated from Eqs. (25a) or (25b).

The inverse Langevin function, needed to calculate α (Eq. (26)) does not have an analytical solution. Numerical solutions can be used to evaluate $\mathscr{L}^{-1}(m)$, which can be calculated to any precision. However, simpler and more efficient analytic formulae are usually preferred over numerical methods [43–45]. In the last few years, several analytical approximants of the inverse Langevin function have been developed, evidencing the current interest and ongoing advances on the subject [44]. In this work, we used the following expression reported by Kröger, due to its simultaneous simplicity and accuracy (maximum relative error < 0.3 % [46]):

$$\mathscr{L}^{-1}(m) = h \cong \frac{3m-\frac{m}{5}(6m^2+m^4-2m^6)}{1-m^2} \tag{27}$$

Both the Langevin function and Kröger's approximant are shown in Fig. 1, together with the relative errors of the approximation for constant

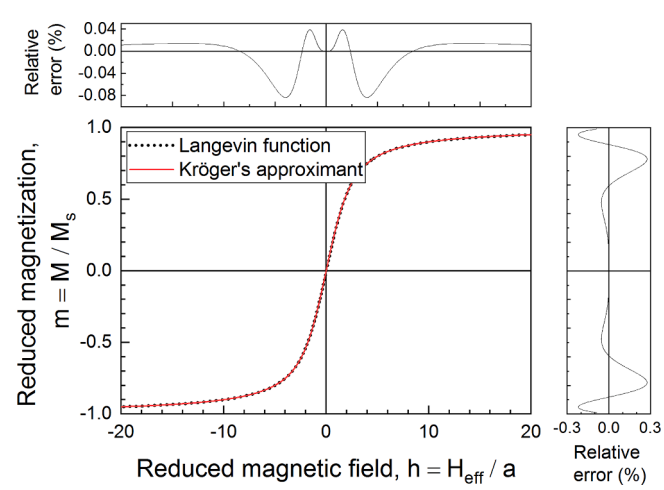

Fig. 1. Dependence of the reduced magnetization ($m = M/M_S$) on the reduced magnetic field ($h = \frac{H_{eff}}{a}$), according to the Langevin function ($\mathscr{L}(h) = \coth(h) - 1/h$) and to Kröger's approximant of the inverse Langevin function ($\mathscr{L}^{-1}(m) \cong \frac{3m-\frac{m}{5}(6m^2+m^4-2m^6)}{1-m^2}$ [46]). In the upper and left plots are the relative errors of the approximation for constant h and m, respectively.

h and m.

Let us now analyze the sign of α that arises from Eq. (26). The coefficient $\frac{H_{TIP}}{M_{TIP}}$ will always be positive, given that the anhysteretic curve lies on the first and third quadrants of the $M-H$ plane. If the $m(h)$ function were linear, then $\frac{m_X}{m_{TIP}}$ would be equal to $\frac{h_X}{h_{TIP}}$. But as $m(h)$ is concave down in the first quadrant (Fig. 1), $\frac{m_X}{m_{TIP}} > \frac{h_X}{h_{TIP}}$ and hence the denominator of the fraction must be positive as well. At last, we notice that $\frac{h_X}{h_{TIP}}$ may be either smaller or larger than $\frac{H_X}{H_{TIP}}$, depending on the shape of the curve, and that this determines the sign of α:

$$\mathrm{sgn}(\alpha) = \mathrm{sgn}\left(\frac{h_X}{h_{TIP}} - \frac{H_X}{H_{TIP}}\right) \tag{28}$$

The above dependence is illustrated in Fig. 2: the more negative the molecular constant is, the wider the range in which the magnetization $M(H)$ is linear, i.e., the wider the range in which the susceptibility $\chi(H)$ is uniform and the higher the anisotropy field is (for a study on the magnetization curve of materials exhibiting induced transverse anisotropy, see [47]).

Step 6: For the chosen m_{TIP} and M_X values, compute the normalized root-mean-square error (NRMSE) of the anhysteretic magnetization fittings, which is the objective function of the optimization problem to minimize:

$$NRMSE = \frac{\sqrt{\sum_{i=0}^{TIP}\left(H_i^{data}\left(M_i^{data}\right) - H_i^{sim}\left(M_i^{data}\right)\right)^2}}{H_{TIP}^{data}\left(M_{TIP}^{data}\right)} \tag{29}$$

being,

$$H_i^{sim}\left(M_i^{data}\right) = a\mathscr{L}^{-1}\left(\frac{M_i^{data}}{M_S}\right) - \alpha M_i^{data} \tag{30}$$

To evaluate the latter equation, which results from solving Eq. (17) for H, the previously calculated parameters must be used (M_S in Eq. (22), α in Eq. (26), and a in Eq. (25)), while the Langevin inverse function can be computed using Kröger's approximation (Eq. (27)). Notice that, although it is common to compute the root-mean-square error with the differences between the magnetizations ($M_i^{data}\left(H_i^{data}\right) - M_i^{sim}\left(H_i^{data}\right)$) [48] or the induction fields ($B_i^{data}\left(H_i^{data}\right) - B_i^{sim}\left(H_i^{data}\right)$) [7], we opt to use the magnetic fields ($H_i^{data}\left(M_i^{data}\right) - H_i^{sim}\left(M_i^{data}\right)$). The reason is that calculating $M_i^{sim}\left(H_i^{data}\right)$ requires to solve Eq. (17) through an iterative numerical method, whereas $H_i^{sim}\left(M_i^{data}\right)$ can be directly determined by Eq. (30) –using an approximant of the inverse Langevin function.

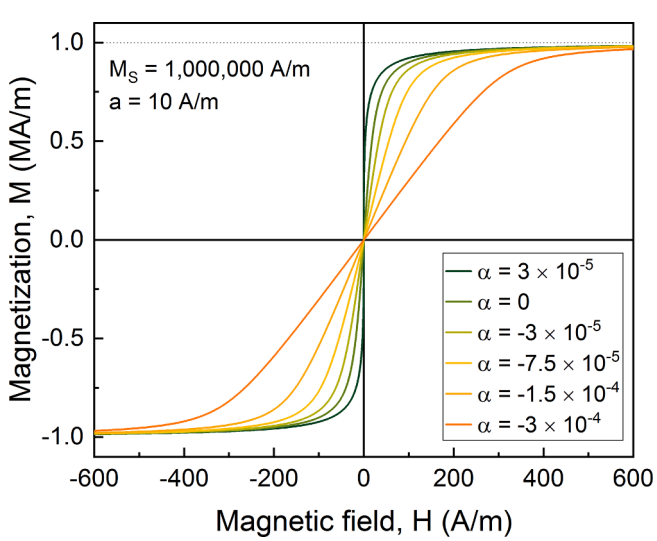

Fig. 2. Anhysteretic magnetization curves according the Langevin-Weiss equation of state with $M_S = 1,000,000\,A/m$, $a = 10\,A/m$, and α ranging from 3×10^{-5} (i.e., $3a/M_S$) to -3×10^{-4}.

Step 7: Explore the search domain ($0 < m_{TIP} < 1$, $0 < M_X < M_{TIP}$) with a nonlinear solving method (e.g. the generalized reduced gradient nonlinear solver) and locate the best solution, which minimizes the objective function. We will show in the next section that this is a smooth and well-behaved optimization problem. Thus, a good and faster solution can be reached by strategically fixing the second constraint to be, for example, $M_X \sim 0.7M_{TIP}$.

5. Experimental and methods

Next, we will validate our proposed method to retrieve the Langevin-Weiss model's parameters for three widely used soft magnetic materials: i) a grain-oriented silicon steel (M130-27 s) measured parallel to the rolling direction, ii) a soft ferrite ($Mn_{0.51}Zn_{0.44}Fe_{2.05}O_4$) with very low anisotropy, and iii) an iron-based nanocrystalline alloy ($Fe_{73.5}Cu_1Nb_3Si_{15.5}B_7$, under the trademark NANOPERM LM from Magnetec Company) with induced in-plane uniaxial anisotropy, measured transverse to the preferred direction.

Grain-oriented silicon steels are the materials of choice of large-scale transformers. The addition of silicon to soft iron results in a significant decrease in coercivity and magnetostriction, and an increase in electrical resistivity, bringing down the power losses. Then, a controlled cold rolling process promotes grain growth along a crystalline direction of low anisotropy, increasing the magnetic susceptibility in the rolling direction even further at a relatively low cost [1].

Soft ferrites stand out from other magnetic materials for high frequency applications. Ferrites are semiconductors with an electrical resistivity six orders of magnitude higher than Fe-based metallic alloys and therefore suffer from low eddy current losses. In contrast to the other soft magnetic materials analyzed here, soft ferrites are not ferromagnetic but ferrimagnetic in nature, and that is why their saturation magnetization is relatively low [1].

Fe-based nanocrystalline alloys are produced in the form of ribbons, which can be wound into cores. Their unique nanostructure and thin thickness work together to keep hysteresis and eddy-current losses low at a reasonable saturation magnetization. Furthermore, the type of the $M(H)$ curve squareness can be controlled by either a magnetic field or a tensile stress annealing treatment, as both yield a uniform uniaxial induced anisotropy [49]. Here, we will fit the anhysteretic magnetization of a nanocrystalline toroidal core with strong induced transverse anisotropy, which is the most advantageous material for current transformers [50].

In the absence of a direct measurement of the anhysteretic magnetization, the curve may be approximated as the average curve of the ascending and descending branches of the major hysteresis loop, as already described in section "1. Introduction" [11,12]. Experimental measurements of quasistatic hysteresis loops were acquired by the fluxmetric method and freely distributed by Szewczyk through his GitHub repository [51]. Measurements were carried out with a magnetizing field frequency of 0.5 Hz at room temperature using a digitally controlled hysteresisgraph. The Epstein test frame was used for the steel, while toroidal cores for the ferrite and nanocrystalline samples (further measurement details reported elsewhere [7,52,53]). The data tables, originally provided as magnetic induction B vs applied field H, were converted into M vs H through the relation $M = \frac{B}{\mu_0} - H$. Then, for a given value of M, we estimated the corresponding anhysteretic value of H as the average of the two hysteretic H values, one for each branch (as opposed to calculating, for a given value of H, the average value of M for both branches, a method described by Bozorth [2] to obtain an approximation of the "normal magnetization curve", i.e., the first magnetization curve). Fig. 3 reports both the experimental hysteresis loops and the calculated anhysteretic magnetization curves of the three soft magnets.

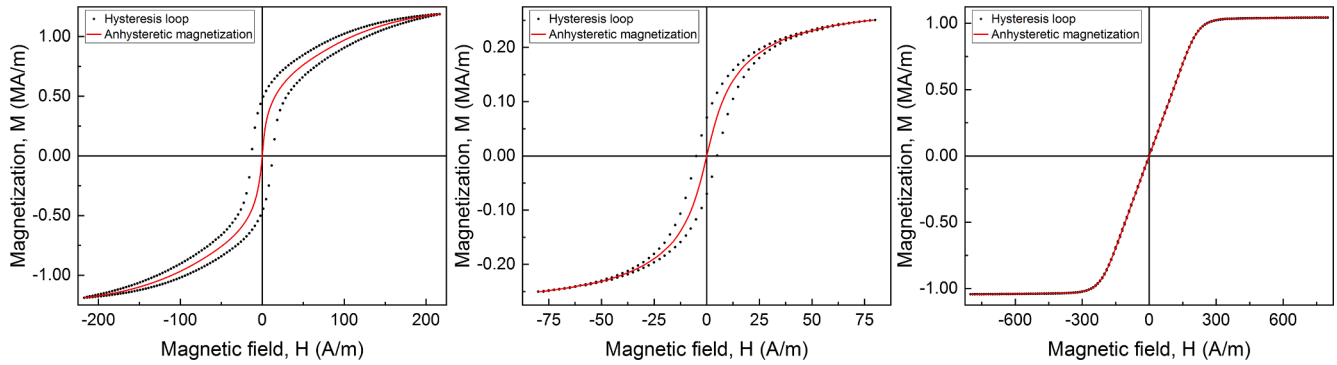

Fig. 3. Black dotted curves show the quasistatic hysteresis loops as measured for different materials [51]. Left: M130-27 s grain-oriented silicon steel, measured parallel to the rolling direction. Center: Mn-Zn ferrite ($Mn_{0.51}Zn_{0.44}Fe_{2.05}O_4$). Right: NANOPERM LM nanocrystalline alloy ($Fe_{73.5}Cu_1Nb_3Si_{15.5}B_7$) with induced in-plane uniaxial anisotropy, measured transverse to the preferred direction. Red solid curves are the corresponding anhysteretic magnetizations, calculated as the average curve of the upper and lower branches of the hysteresis loop.

6. Results and discussion

As commented before, the optimum values of m_{TIP} and M_X that allow calculating the Langevin-Weiss model parameters –M_S, α, and a– are obtained by finding the global minimum of the objective function; the normalized root-mean-square error (see Step 7 in section "4. A new method for the determination of the Langevin-Weiss model parameters"). In Fig. 4 we show that the objective function has a smooth shape for all three cases. The best solution, pointed out with a white star in each contour plot, could be located searching throughout the bounded space domain: $0 < m_{TIP} < 1$, $0 < M_X < M_{TIP}$. But as the objective function has a small dependence on the second constraint, strategically setting $M_X \cong 0.7M_{TIP}$ and solving for the optimum m_{TIP} only, a faster and still good solution could have been reached.

Once the best-fit values of m_{TIP} and M_X were determined, M_S, α, and a were calculated from Eqs. (22), (26), (25), respectively. Fig. 5 illustrates that the three modelled curves are in good agreement with the experimental anhysteretic magnetizations. Yet, we believe that the more homogeneous nature of ferrite and nanocrystalline materials, as compared to that of steels, led to a better accuracy of the model in those cases.

The method for finding the Langevin-Weiss function's parameters presented here can easily be extended to consider the higher order terms of the molecular field Weiss suggested for some cases (Eq. (15)). To this end, we still need to propose m_{TIP} and determine M_S (steps 1 and 2 of our method). The number of magnetization values we need to choose (step 3), equals the number of the molecular field terms (e.g., M_{X1} and M_{X2} when $H_{int} = \alpha_1 M + \alpha_3 M^3$). The corresponding magnetic fields (e.g., H_{X1} and H_{X2}), should again be estimated by interpolation of the data curve (step 4). Then (step 5), the $1+n$ unknowns (a, α_1, α_3, $\cdots$) can be solved from the following system of $1+n$ linear equations that arise from evaluating the Langevin-Weiss function solved for H at the tip of the curve (i.e. (H_{TIP}, M_{TIP})) plus the other n points strategically selected (i.e. $(H_{X1}, M_{X1}), (H_{X2}, M_{X2}), \cdots$):

$$-a\mathscr{L}^{-1}(M_{TIP}/M_S) + \alpha_1 M_{TIP} + \alpha_3 M_{TIP}^3 + \cdots = -H_{TIP} \tag{31a}$$

$$-a\mathscr{L}^{-1}(M_{X_n}/M_S) + \alpha_1 M_{X_n} + \alpha_3 M_{X_n}^3 + \cdots = -H_{X_n} \tag{31b}$$

The latter can be solved for the a parameter and the molecular field constants through the method of Gaussian elimination and back substitution (both algorithms can be found in [54]). Finally, as previously described, m_{TIP} should be varied (between zero and unity) until the normalized root-mean-squared error is minimized (steps 6 and 7). Here, we only report the results obtained with the first order term of the molecular field because the NRMSE did not significantly improve, for any of the fitted curves, when we included up to either the third or fifth order terms.

Notice that the molecular field resulted to be negative for the nanocrystalline material with strong induced anisotropy, where the magnetization had been measured along the material's hard axis. This may seem physically counterintuitive as ferromagnetism is a cooperative phenomenon. A possible explanation is that the reduced magnetic field, i.e. the argument of the Langevin function, is still enhanced by the interplay of domain coupling, magnetocrystalline, and magnetoelastic anisotropies, since the $a = \frac{Nk_BT}{\mu_0 M_S}$ parameter is much smaller than in paramagnets, where these effects do not exist. Actually, it is well known that magnetic crystal forces may oppose the action of the applied field

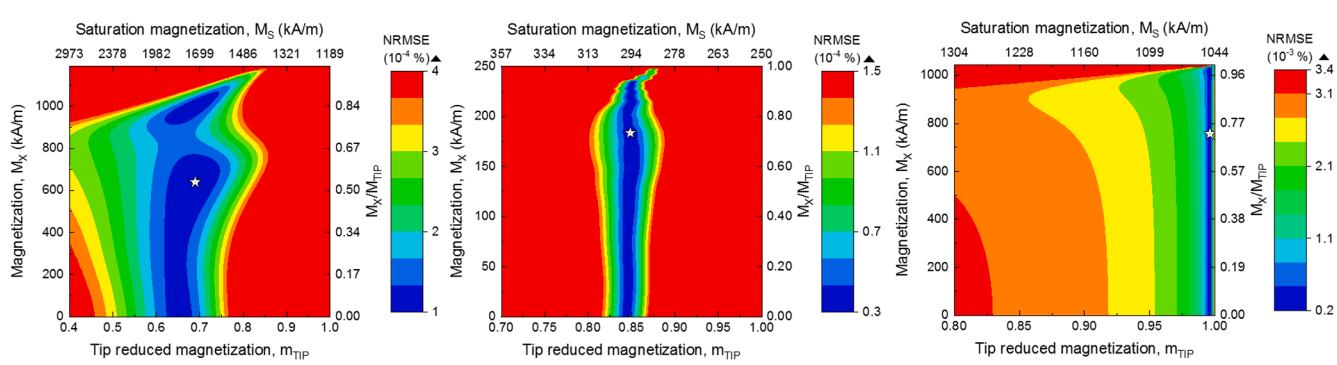

Fig. 4. Contour plots showing the shape of the objective function; the normalized root-mean-square error (NRMSE) of the anhysteretic magnetization fittings for the M130-27 s grain-oriented silicon steel (left), Mn-Zn ferrite (center), and NANOPERM LM nanocrystalline alloy (right). The lower horizontal axis is the reduced magnetization at the tip of the anhysteretic curve (m_{TIP}), while the upper horizontal axis shows the corresponding saturation magnetization value ($M_S = m_{TIP} \times M_{TIP}$). The vertical axis is the magnetization field at the second constraint point of the anhysteretic curve (M_X). The stars point out the best-fit values of m_{TIP} in each case.

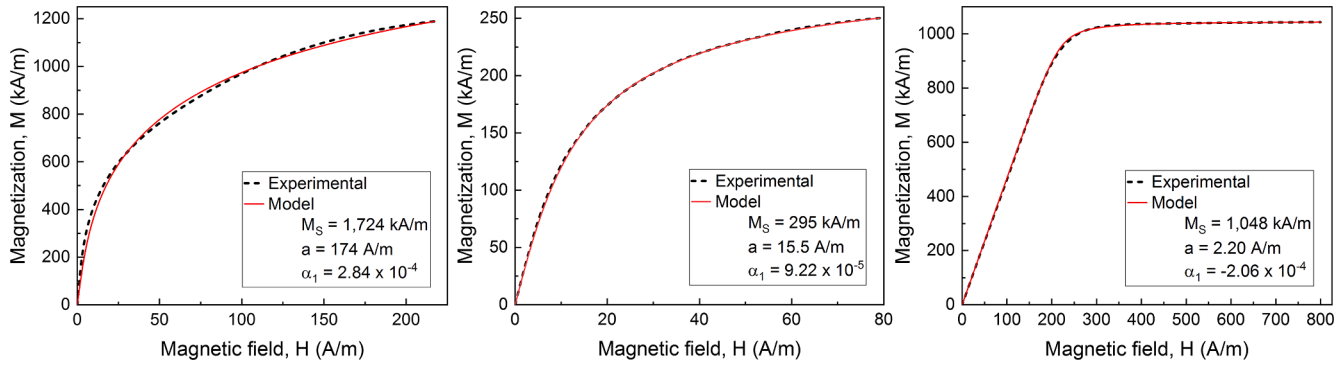

Fig. 5. Experimental anhysteretic magnetization curves (black dotted lines, calculated as shown in Fig. 3) fitted with our method based on the Langevin-Weiss model (red solid lines, with the constrain values optimized as shown in Fig. 4) for the M130-27 s grain-oriented silicon steel (left), Mn-Zn ferrite (center), and NANOPERM LM nanocrystalline alloy (right). The corresponding model parameters are reported in each plot.

[2]. Also, as mentioned in section "1. Introduction", although the exchange integral J_0 and thus the molecular field constant α are commonly accepted to be positive for ferromagnets, no theoretical demonstration has been provided yet for real materials.

7. Conclusions

After a brief overview of the anhysteretic magnetization of soft magnets, we have demonstrated that the Langevin-Weiss theory is successful in describing the magnetic field dependence of the anhysteretic magnetization of any soft ferro and ferrimagnetic material with an approximately homogeneous distribution of magnetic moments.

We presented a computationally efficient and yet accurate deterministic method for retrieving the Langevin-Weiss model's parameters even for scenarios commonly thought as challenging: curves with high initial susceptibilities and curves measured transversely to the direction of strong induced anisotropies. The two variables of the proposed optimization problem (m_{TIP} and M_X) are intrinsically bounded, as compared to the variables of the typical approach: neither the upper bounds of M_S, a, nor α are obvious and even the nonnegative lower bound of α may be disputed. In fact, negative molecular field constants should be expected when strong anisotropies have been induced transversely to the measurement direction of the material. Further studies are needed to explore the relation between the Langevin-Weiss model's parameters and the direction of the applied field with respect to the easy axis in anisotropic materials.

Our strategy has been validated for three widely used soft magnetic materials: a grain-oriented silicon steel measured along its preferred axis, a Mn-Zn soft ferrite, and an iron-based nanocrystalline alloy with transverse induced anisotropy.

The proposed method is attractive from the engineering point of view as it enables an easy and reliable modeling of the nonlinear behavior of homogeneous soft magnetic components. Additionally, even the popular Jiles-Atherton model for ferromagnetic hysteresis may benefit from this method, as some researchers have recommended to first find the anhysteretic magnetization parameters (M_S, a, and α) to then fit the other parameters (k and c) in a subsequent fitting step [52,55].

CRediT authorship contribution statement

Josefina María Silveyra: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Juan Manuel Conde Garrido:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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